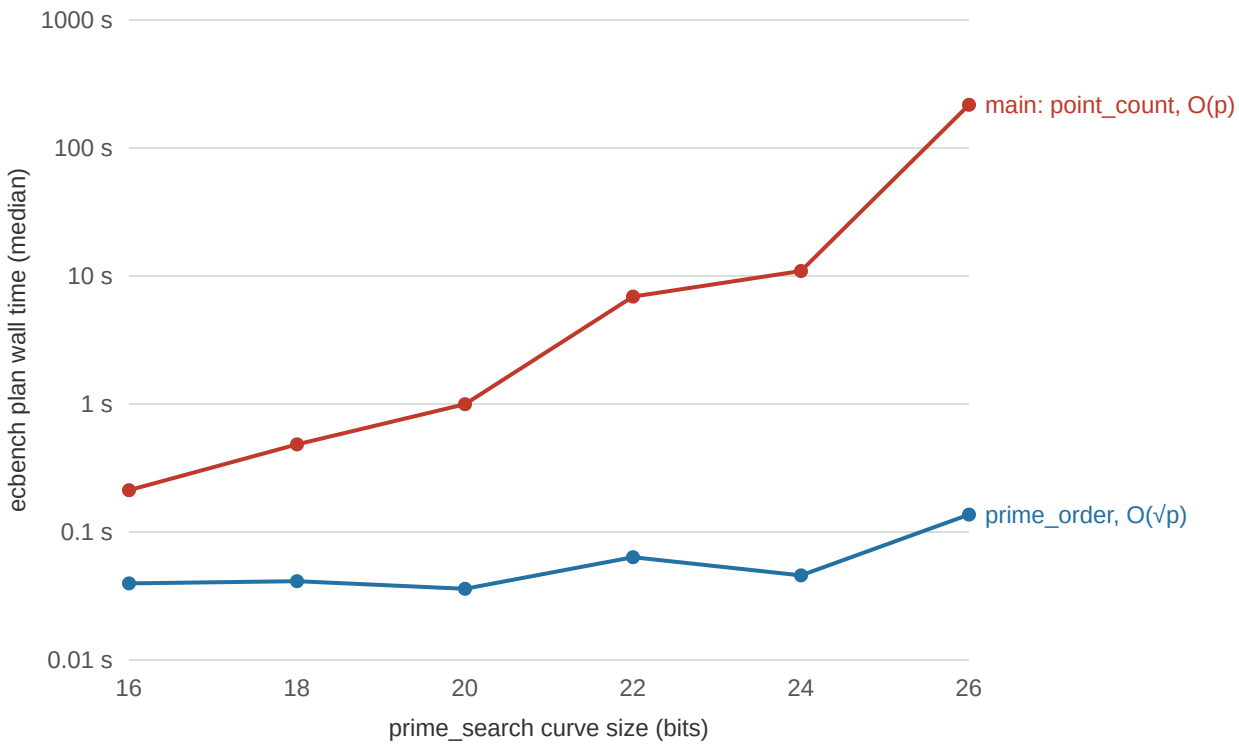
Prime-order curve search: O(p) → O(√p)

Wall time of `ecbench plan` on a one-curve `prime_search` spec (seed 59297), median of 3 runs (one run for main at 26 bits). The plan is dominated by `find_prime_order_curve`.



bits	main (s)	this change (s)	speedup
16	0.212	0.040	5×
18	0.484	0.041	12×
20	0.997	0.036	28×
22	6.901	0.064	109×
24	10.956	0.046	239×
26	217.607	0.137	1592×

Correctness: the search draws the same random numbers and accepts the same curves. All 21 committed `prime_search` sessions on main re-plan to their recorded workload IDs (288 workloads, `replan-check.txt`); `prime_order` equals point_count + primality on 400 random curves, and the search equals the naive search at 8–16 bits for three seeds (unit test `prime_order_matches_the_naive_count`).

Host: Intel Xeon @ 2.10 GHz, 4 vCPU cloud container, shared, unpinned, L0 wall. Engineering change to setup time; no solve, cost unit or measured result changes.